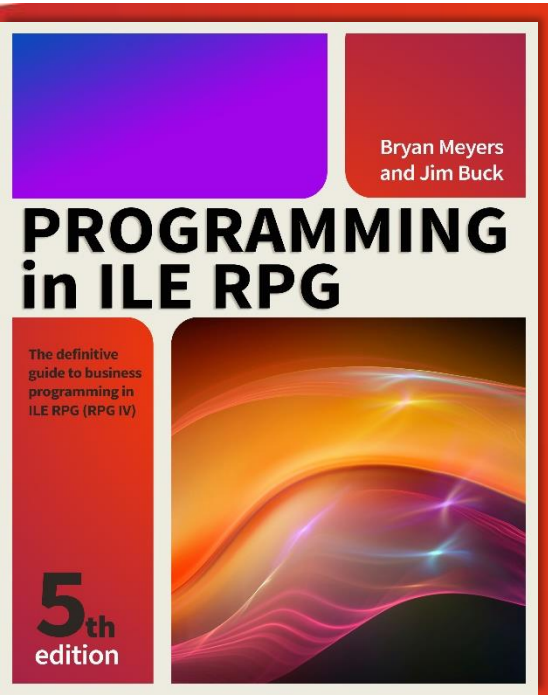
Chapter 1:Introduction to Programming and RPG



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A note on using these PowerPoints

- Some slides have sound clips, denoted by a speaker symbol that you will see on some of the slides. Please listen to these sound clips.
- Some slides also have links to video clips, denoted by “watch this clip”
- Slide designs have been removing from everything except the opening slide to save on toner, ink, etc. should you choose to do any printing.

History of RPG

- Report Program Generator
  - Introduced in 1960s
- RPG II
  - Columnar fixed format code
  - RPG cycle
  - Program described files
  - Heavy use of numbered indicators
  - System/3, System/32, System/34, System/36 midrange computers



```
FINVHDR  IP  F      62            DISK
FPRINTER O   F     132     OF     PRINTER
IINVHDR  NS  01    1NC
I                                       1   3 STORE L1
I                                       4  13 INVNO
I                                      14  20 CUSTNO
I                                      21  45 STNAM
I                                      46  53 INVDAT
I                                      54  622TOTINV
I           UDS
I                                       1   8 RPTDAT
C   01      INVDAT    COMP RPTDAT                   11
C   01 11             ADD  TOTINV    L1TOT   92
C   01 11             ADD  TOTINV    LRTOT   92
OPRINTER H  101   1P
O       OR        OF
O                         PAGE  Z  106
O                                 102 'PAGE'
O                                  59 'VERY BIG'
O                                  72 'STORES, INC.'
O                         UDATE Y   17
O                                   8 'RUN DATE'
O        H  1     1P
O       OR        OF
O                                  73 'DAILY SALES BY STORE'
O        H  2     1P
O       OR        OF
O                         RPTDAT    83
O                                  63 'FOR DATE:'
```

History of RPG

- RPG III
  - Externally described files
  - Structured design concepts
  - Interactive applications
  - System/38 introduced in 1979
  - AS/400 (RPG/400) introduced in 1988



History of RPG

- ILE RPG (RPG IV)
  - Integrated Language Environment (ILE)
  - Introduced in 1994
  - Modular programming, procedures (part of Advanced RPG course)
  - Service programs (part of Advanced RPG course)
  - Free format
    - In 2001, the calculations went “free form” but the rest of the specs were still fixed form
    - In 2014, RPG IV became “all free” for ALL specs
  - Expanded name lengths, data types, limits
  - Runs under IBM i operating system

What is a Program Variable?

- Named data item that represents memory location
  - Allow processing by name rather than by memory address
- Sometimes called “field”
  - “Field” reserved for variable that reside in a file record layout
- Variables and fields are strictly typed
  - Must be defined to program with specific data type and length (and decimal positions, if numeric)

What is an Object?

- Almost every named IBM i entity is an object
  - Categorized by type
  - Object type identifies purpose of object
- Library (*LIB)
  - Container for other objects
- File (*FILE)
  - Holds formatted data records
- Program (*PGM)
  - Contains executable code
- Command (*CMD)
  - Shortcut to execute program

Watch this clip:
https://youtu.be/3LVI97w0iFE

Object Naming

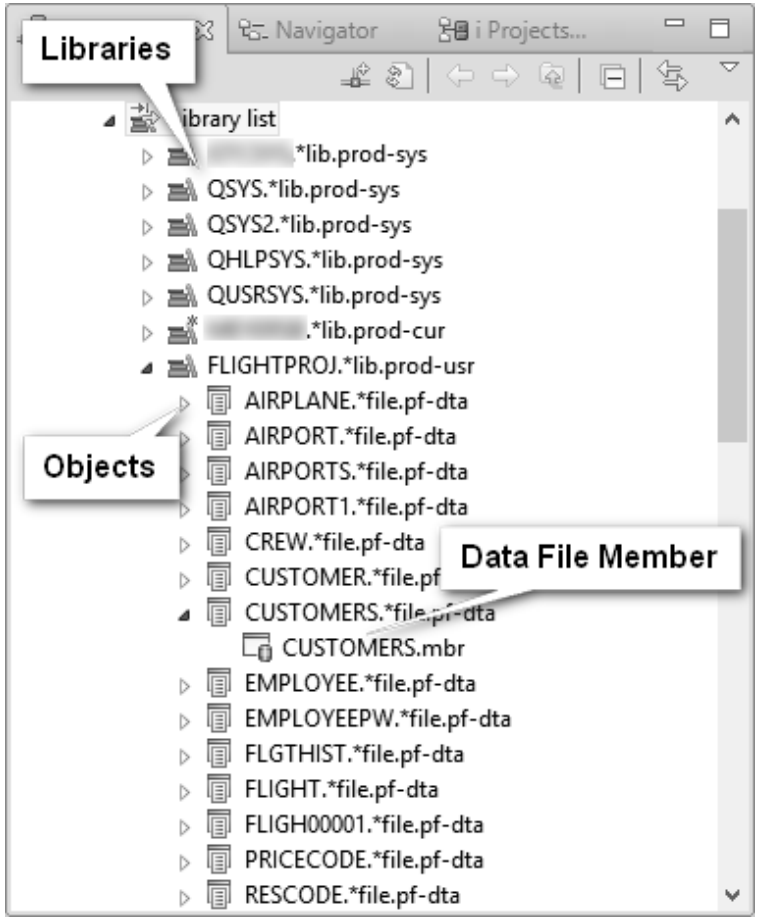
- May refer to simple object name
  - System will search library list path to find object
  - E.g., CUSTOMERS
- May qualify name with library name
  - System will find object in specified library
  - E.g., FLIGHTPROJ/CUSTOMERS

Database Files

- Physical file (*FILE PF-DTA)
  - Contains data
  - SQL table
- Logical file (*FILE LF)
  - Customized view of data in physical file
  - Provides sorting and selection criteria for processing file data
  - Access path is maintained dynamically
  - SQL view
- Source physical file (*FILE PF-SRC)
  - Specially formatted physical file for program source statements

Database File Members

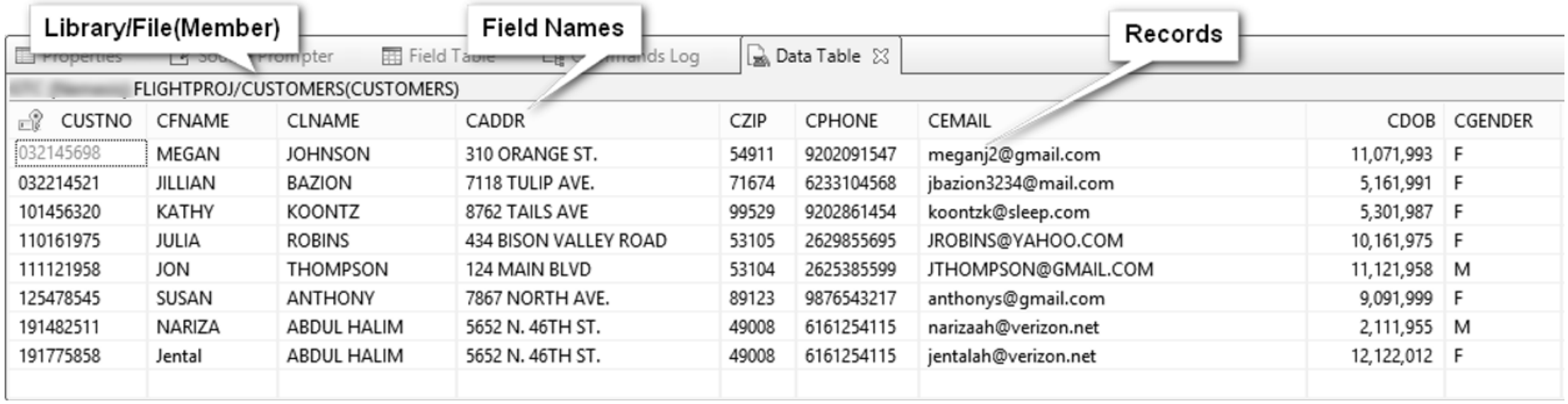
- Member is a subset of data in a file
  - Every file has at least one member
  - Files may contain multiple members
- Program typically processes only one member
- Source files use multiple members to store source statements for multiple programs



Data File Organization

- File
  - Collection of data about kind of entity (“table”)
  - E.g., Customers
- Member
  - Subset of file (“data set”)
- Record
  - Data about one instance of entity (“row”)
  - E.g., Customer# 032145698
- Field
  - Discrete formatted piece of information about each record (“column”)
  - Organized into named record format
    - One format per physical file
  - E.g., Customer#, Name, Address, Phone#

Data File Organization



CUSTNO	CFNAME	CLNAME	CADDR	CZIP	CPHONE	CEMAIL	CDOB	CGENDER
032145698	MEGAN	JOHNSON	310 ORANGE ST.	54911	9202091547	meganj2@gmail.com	11,071,993	F
032214521	JILLIAN	BAZION	7118 TULIP AVE.	71674	6233104568	jbazion3234@mail.com	5,161,991	F
101456320	KATHY	KOONTZ	8762 TAILS AVE	99529	9202861454	koontzk@sleep.com	5,301,987	F
110161975	JULIA	ROBINS	434 BISON VALLEY ROAD	53105	2629855695	JROBINS@YAHOO.COM	10,161,975	F
111121958	JON	THOMPSON	124 MAIN BLVD	53104	2625385599	JTHOMPSON@GMAIL.COM	11,121,958	M
125478545	SUSAN	ANTHONY	7867 NORTH AVE.	89123	9876543217	anthonys@gmail.com	9,091,999	F
191482511	NARIZA	ABDUL HALIM	5652 N. 46TH ST.	49008	6161254115	narizaah@verizon.net	2,111,955	M
191775858	Jental	ABDUL HALIM	5652 N. 46TH ST.	49008	6161254115	jentalah@verizon.net	12,122,012	F

Record Layout

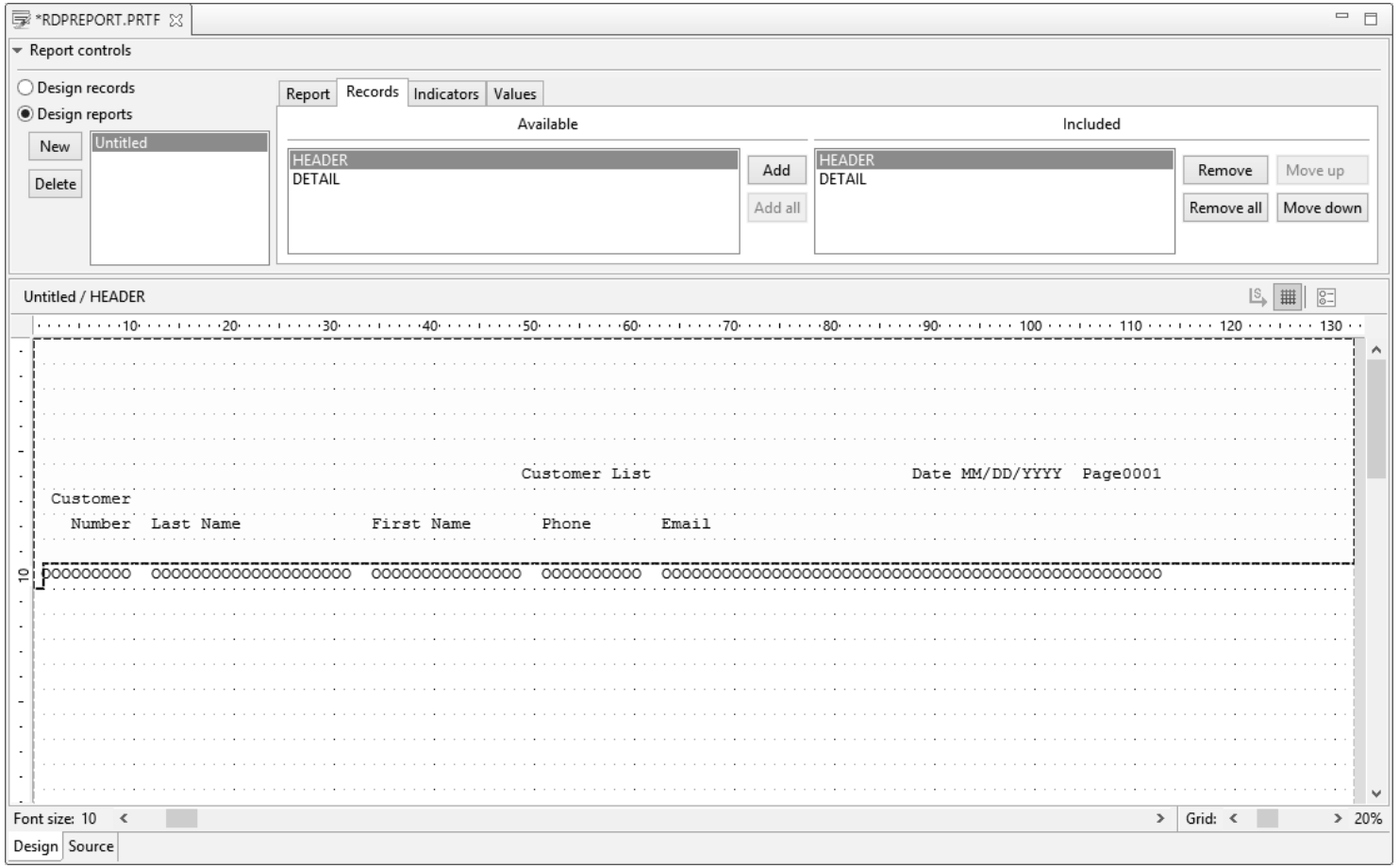
Properties | Source Prompter | Field Table | Commands Log | Data Table

[redacted]: File FLIGHTPROJ/CUSTOMERS (9 Fields)

Name	Record	Type	Length	Text
CUSTNO	CUSTSREC	Character	9	CUSTOMER NUMBER
CFNAME	CUSTSREC	Character	15	CUSTOMER FIRST NAME
CLNAME	CUSTSREC	Character	20	CUSTOMER LAST NAME
CADDR	CUSTSREC	Character	30	CUSTOMER STREET ADDRESS
CZIP	CUSTSREC	Character	5	ZIP CODE
CPHONE	CUSTSREC	Character	10	CUSTOMER PHONE
CEMAIL	CUSTSREC	Character	50	CUSTOMER EMAIL
CDOB	CUSTSREC	Packed Decimal	8.0	CUSTOMER DATE OF BIRTH
CGENDER	CUSTSREC	Character	1	F=FEMALE M=MALE



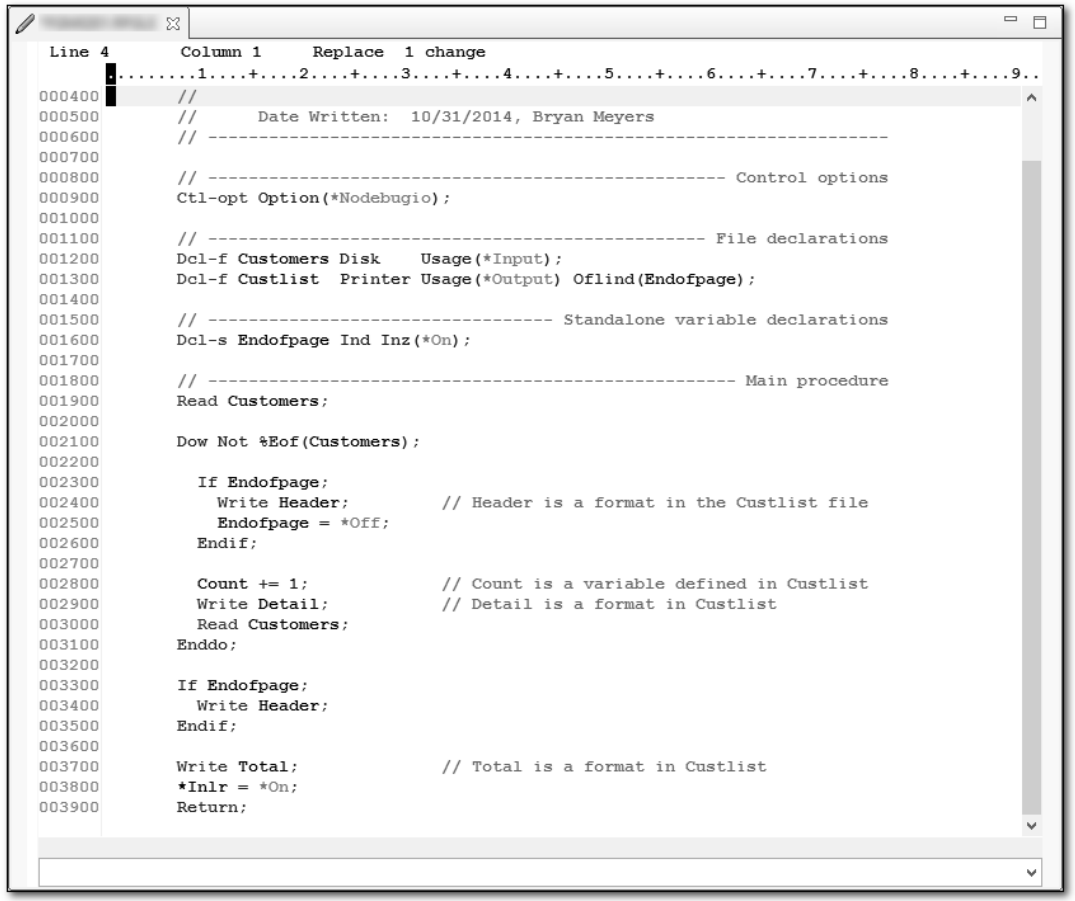
Report Designer – part of “RDi” used later in course (listen to audio clip)



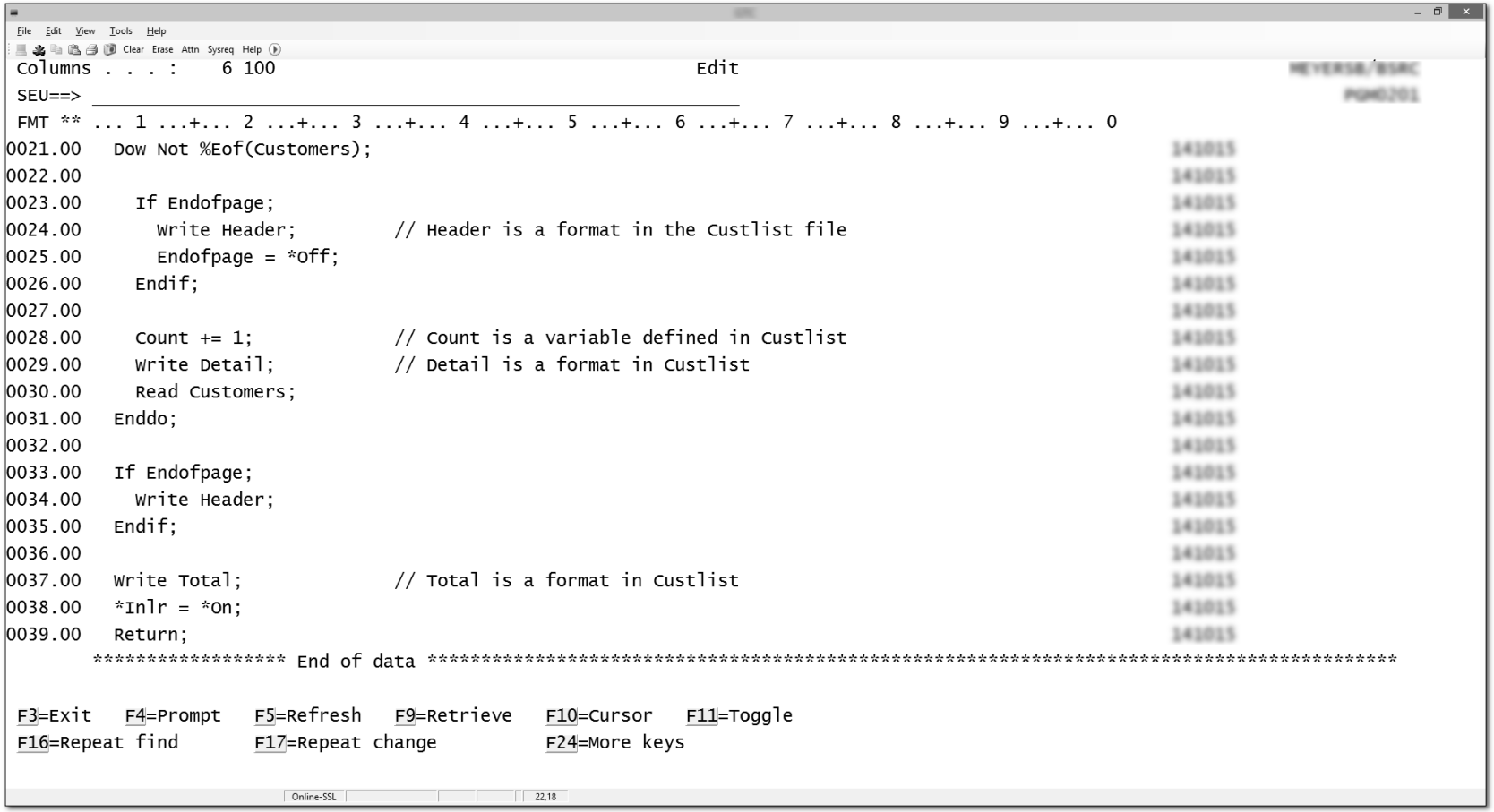
Integrated Development Environments

- Rational Developer for i (RDi) – MORE ON THIS LATER
  - Graphic PC client development environment
  - Based on open source Eclipse platform
  - Live Parsing Extensible (LPEX) editor
- Programming Development Manager (PDM) – OUR FOCUS TO START
  - MORE “How to” information next week
  - 5250 “Green screen” server-based development environment
  - Named “Green Screen” because of the appearance of the screen
    - Can be changed to other colors
  - Source Entry Utility (SEU) editor

Live Parsing Extensible (LPEX) Editor
Part of “RDi” – more on this later



Source Entry Utility (SEU) – more next week



Program Entry and Testing

